SEMESTER 1 • ENGINEERING SCIENCE • PRACTICAL DRAWING HANDBOOK

Engineering Graphics

A complete student-friendly digital textbook for engineering drawing, geometrical constructions, projections, orthographic and sectional views, isometric drawing, freehand sketching, and CADD basics.

COURSE CODE **1005**	COURSE TITLE **Engineering Graphics**
PROGRAMME **Diploma in Engineering and Technology**	SEMESTER **1**
CREDITS **1.5**	CATEGORY **Engineering Science**
PERIODS / WEEK **3 (L:0, T:0, P:3)**	PERIODS / SEMESTER **45**

45

Periods of drawing practice

Engineering Graphics is a technical language. This handbook explains the rule, shows construction steps, gives practice and complete answer procedures.

COURSE OVERVIEW

What is Engineering Graphics?

Engineering Graphics is the language used by engineers, technicians, designers and workshop teams to communicate shape, size, location, construction method and manufacturing details. A drawing can tell a machinist where to cut, a fabricator where to drill, or a maintenance team how to identify a part.

Technical language

Lines, symbols, dimensions and views communicate engineering information without long explanation.

ഡ്രോയിംഗ് ഒരു technical language ആണ്. വര, അളവ്, ചിഹ്നം എന്നിവ ഉപയോഗിച്ച് ആശയം കൃത്യമായി പറയാം.

Visualisation skill

You learn to imagine a 3D object from 2D views and convert real objects into drawings.

2D view കണ്ട് 3D object മനസ്സിൽ കാണാനുള്ള കഴിവാണ് ഇവിടെ പ്രധാനമായത്.

Workshop readiness

Correct drawing practice helps in manufacturing, inspection, assembly and maintenance work.

Workshop-ൽ drawing തെറ്റിയാൽ part തെറ്റും. അതുകൊണ്ട് standard drawing practice വളരെ പ്രധാനമാണ്.

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COURSE DETAILS

Objectives, prerequisite, outcomes and CO-PO mapping

Course Objectives

1. To familiarize the language of graphics used to express ideas and convey instructions while carrying out engineering jobs.
2. To familiarize drafting and sketching skills, drawing equipment, and Indian Standards related to engineering drawings.
3. To apply skills to translate ideas into sketches and to draw and read engineering curves, projections, and dimensioning styles.
4. To apply basic skills to develop projection of points and lines.
5. To familiarize the knowledge of orthographic and sectional views of objects.
6. To apply skills to visualize actual objects or parts of objects based on drawings.
7. To experiment with simple commands used for construction of two-dimensional plane figures in CADD.

Prerequisite

Basic Geometry of Secondary School level.

Basic points, lines, angles, circles, triangles, rectangles and geometrical construction are required.

> School-level geometry അറിയാമെങ്കിൽ Engineering Graphics പഠിക്കാൻ നല്ല foundation ലഭിക്കും.

Course Outcomes

CO	Description	Duration	Level	Student meaning
CO1	Illustrate basic elements of Drawing	14 hours	Understanding	Use instruments, line conventions, dimensioning, curves and scales.
CO2	Construct Projections of points and lines	12 hours	Applying	Draw point and line projections using HP, VP and XY.
CO3	Build Orthographic projections and Sectional views of object	9 hours	Applying	Convert pictorial objects into views and draw sections.
CO4	Develop Isometric Projections	8 hours	Applying	Draw isometric views/projections and use CADD basics.
Series Test	Series Test	2 hours	-	Series Test I after Module 2 and Series Test II after Module 4.

CO-PO Mapping

CO1 → PO1

2, moderately mapped.

CO2 → PO1

3, strongly mapped.

CO3 → PO1

3, strongly mapped.

CO4 → PO1

3, strongly mapped.

Mapping scale: 3 - Strongly mapped, 2 - Moderately mapped, 1 - Weakly mapped.

LEARNING ROADMAP

45-period drawing plan

Module	Duration	Topics	Main output	Exam focus
Module 1	14 hours	Instruments, lines, dimensioning, polygons, conics, helix, involute, scales	Neat construction drawings	BIS rules and curves
Module 2	12 hours + Series Test I	Projection theory, quadrants, points, lines, true length	Front and top views	Projection placement
Module 3	9 hours	Orthographic views, first/third angle, sectional views	FV, TV, SV, sections	First angle and hatching
Module 4	8 hours + Series Test II	Isometric view/projection, freehand sketch, CADD	Isometric drawings and CADD workflows	Conversion and commands

Before you start drawing

A good drawing starts before the first line. Sheet fixing, pencil selection, instrument handling, line quality and dimensioning decide professional quality.

Drawing sheet and board

Fix the sheet squarely using clips or tape. Keep the surface clean.

Pencil grades

Use H/2H for construction lines and HB for object lines. Keep pencil sharp.

Mini drafter / T-square

Use for horizontal and vertical lines. Do not freehand instrument lines.

Set squares

Use 45° and 30°-60° set squares for standard inclined lines.

Compass and divider

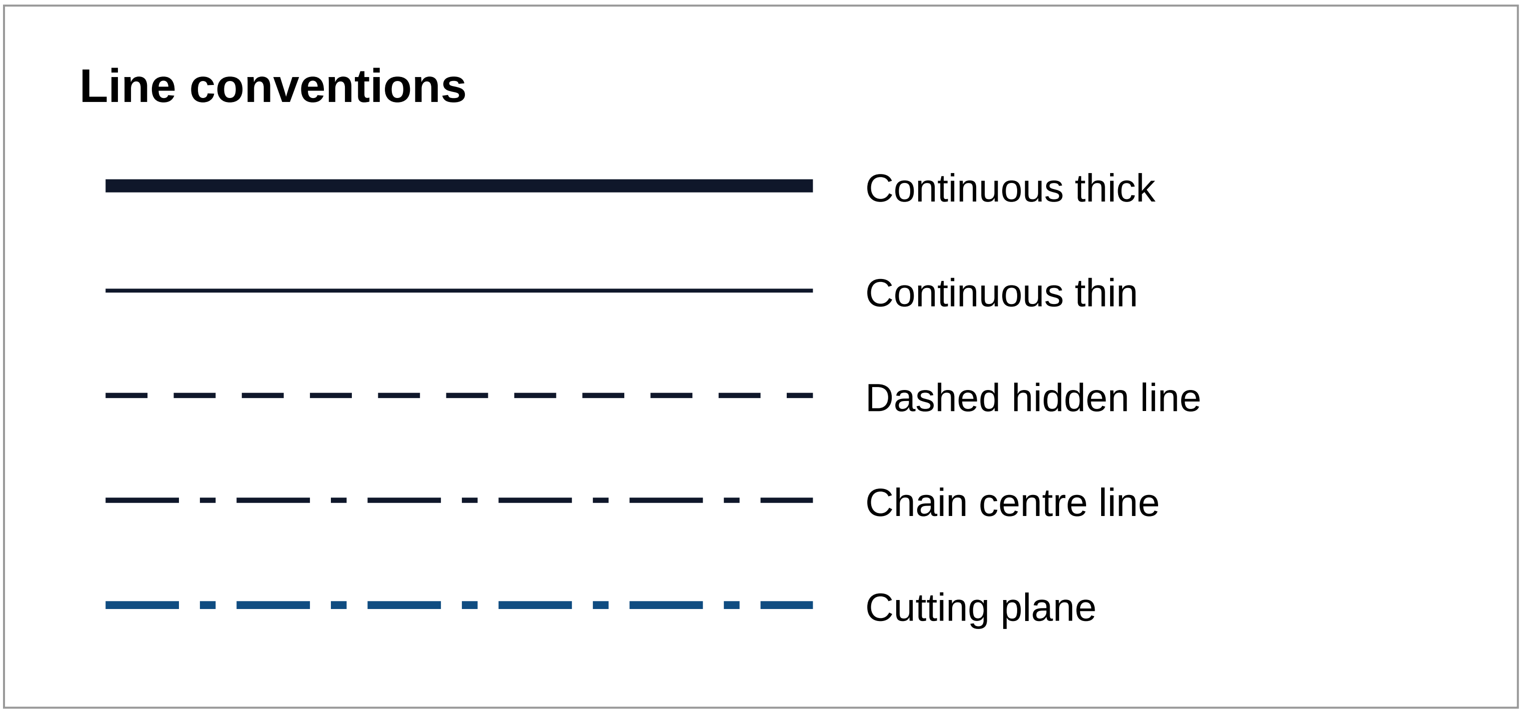
Compass draws arcs/circles. Divider transfers measurements.

Scale and eraser

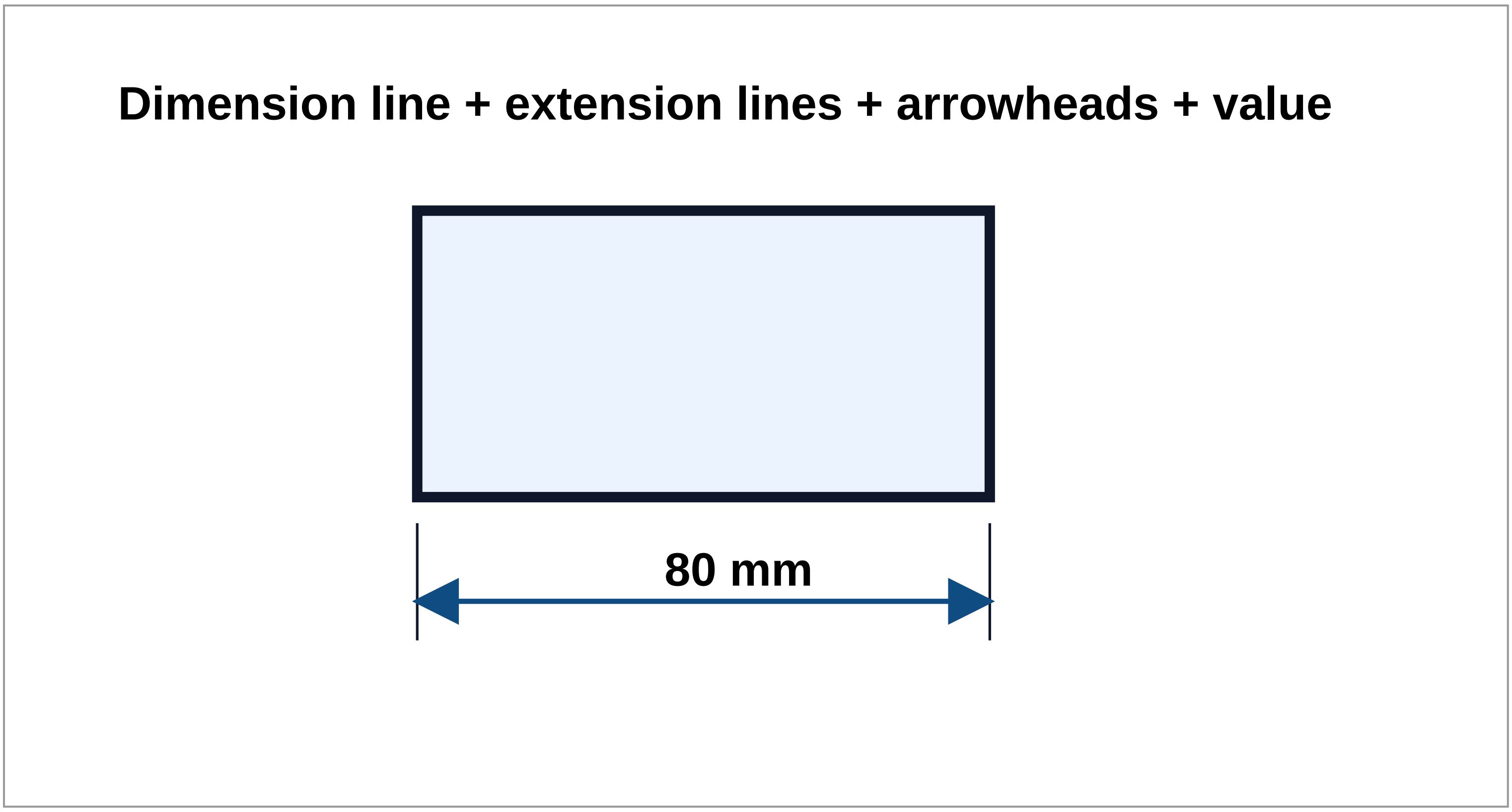
Use scale for measurement, not for heavy pressure. Erase lightly.

Line type	Use	Common mistake
Continuous thick	Visible outlines and edges	Making every line thick
Continuous thin	Dimension, extension, projection and hatching lines	Drawing dimensions too dark
Dashed thin	Hidden edges	Unequal dash spacing
Chain thin	Centre lines and axes	Using solid line for centre line
Cutting plane	Section plane location	Missing arrows and label

Line types diagram



Dimensioning diagram



Common Mistake: Construction lines should be light. They guide the drawing but should not dominate the final answer.

MODULE 1 • CO1

Basic Elements of Drawing

Syllabus focus: drawing instruments, line conventions, BIS dimensioning, chain/parallel/coordinate dimensioning, regular polygons, ellipse, parabola, helix, involute and scales.

Module Outcomes

- M1.01 Outline importance of engineering graphics.
- M1.02 Recognize drawing instruments, standards and symbols.
- M1.03 Appreciate dimensioning.
- M1.04 Develop polygons, conics and engineering curves.
- M1.05 Outline importance of scales.

Engineering drawing as technical language

A drawing communicates shape, size, location and manufacturing information. It follows standards so different people read the same meaning.

ഒരു drawing എല്ലാവർക്കും ഒരേ അർത്ഥത്തിൽ വായിക്കാൻ കഴിയണം. അതിനാണ് standards.

Dimensioning methods

Chain dimensioning

Dimensions are placed end-to-end. Useful for sequential features but error may accumulate.

Parallel dimensioning

All dimensions start from a common reference. It reduces accumulated error.

Coordinate dimensioning

Feature locations are given from a datum using X and Y coordinates.

Regular polygon procedure

1. Draw the given side AB to length.
2. Use suitable geometrical method for the number of sides.
3. Locate centre or required arc points.
4. Mark remaining vertices with compass.
5. Join vertices with continuous thick lines and label.

Conic sections

Conics are curves obtained by cutting a cone with a plane. Important syllabus curves are ellipse and parabola.

Cone ഒരു plane കൊണ്ട് മുറിക്കുമ്പോൾ കിട്ടുന്ന curves ആണ് conic sections.

Ellipse by eccentricity method

1. Draw directrix and axis.
2. Mark focus on the axis.
3. Use eccentricity ratio e = distance from focus / distance from directrix.

4. Locate points satisfying the ratio.
5. Join points smoothly and label.

Ellipse by concentric circle method

1. Draw major and minor axes.
2. Draw two concentric circles with semi-major and semi-minor radii.
3. Divide both circles equally.
4. Project verticals from outer circle and horizontals from inner circle.
5. Join intersection points smoothly.

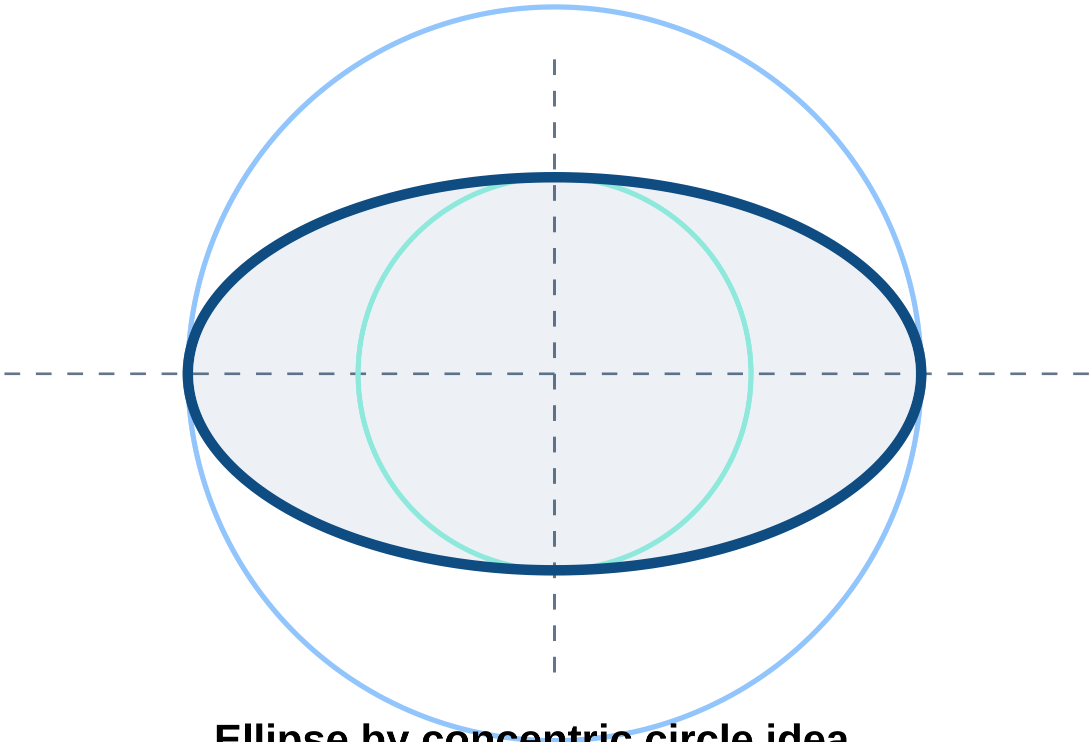
Ellipse by rectangular method

Draw rectangle enclosing major and minor axes, divide sides and axes, project corresponding divisions, locate points and join a smooth ellipse.

Parabola by tangent method

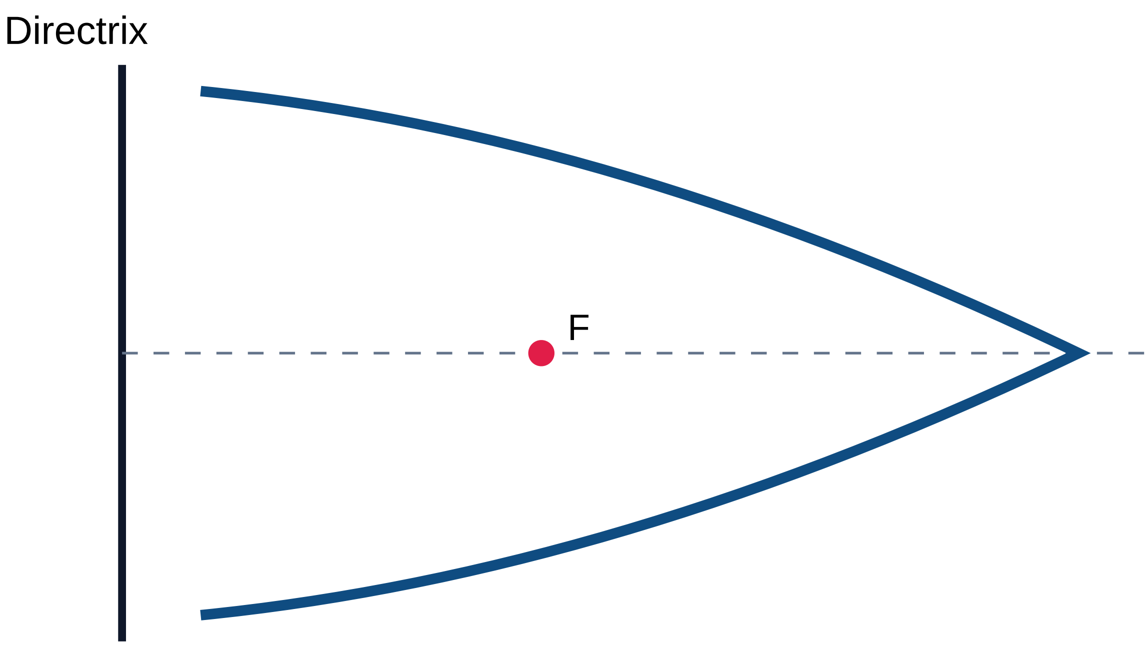
Divide base and side equally, draw tangent construction lines, and draw a smooth curve as the envelope of tangents.

Ellipse diagram



Ellipse by concentric circle idea

Parabola diagram



Parabola focus-directrix relation

Helix

Curve traced by a point moving around a cylinder while moving along its axis uniformly. Examples: screw thread and spring.

1. Draw circle/base and divide equally.
2. Draw development length and pitch height.
3. Divide pitch into same number of parts.
4. Project corresponding divisions and join smoothly.

Involute

Curve traced by the end of a string unwound from a circle. Important in gear tooth profile.

1. Draw base circle and divide equally.
2. Draw tangents at division points.
3. Set off arc lengths on tangents.
4. Join points smoothly to complete involute.

Scales

Scale	Purpose
Plain scale	Measures two units such as metre and decimetre.
Diagonal scale	Measures three units accurately.
Vernier scale	Measures fractional units with greater accuracy.

Solved example: Hexagon side 40 mm

1. Draw AB = 40 mm.
2. With A and B as centres and radius 40 mm, draw arcs meeting at O.
3. Draw a circle with O as centre and radius OA.
4. Step off the same radius around the circle to locate six vertices.
5. Join adjacent vertices and label A, B, C, D, E, F.
6. Keep construction arcs light and final outline dark.

Final drawing: regular hexagon with all sides 40 mm.

Module 1 expected questions

1. Define centre line. Name two dimensioning methods.
2. Construct an ellipse by concentric circle method.
3. Construct a parabola by tangent method.
4. Construct an involute of a circle.
5. Explain plain, diagonal and vernier scales.

MODULE 2 • CO2

Projection of Points and Lines

Syllabus focus: projection theory, points in quadrants, straight lines in first quadrant, true length and inclination. Series Test I follows this module.

Module Outcomes

- M2.01 Outline theory of projection.
- M2.02 Develop projections of points.
- M2.03 Develop projections of lines.

Theory of projection

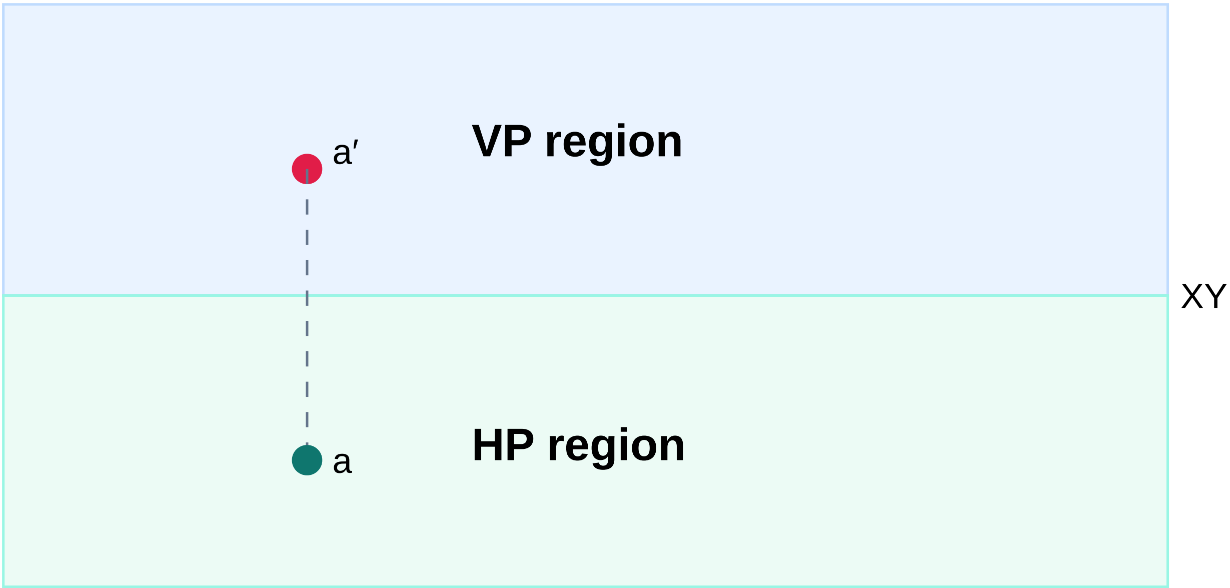
Projection represents a 3D object on a 2D plane using projectors. HP gives top view and VP gives front view.

> 3D object-നെ 2D sheet-ൽ കാണിക്കുന്ന രീതിയാണ് projection. HP, VP ആണ് main planes.

HP, VP and XY

HP: Horizontal Plane. **VP:** Vertical Plane. **XY:** Intersection/reference line between HP and VP.

Point projection diagram



First quadrant: front view above XY, top view below XY

Quadrants

Quadrant	Point position	FV	TV
First	Above HP, in front of VP	Above XY	Below XY
Second	Above HP, behind VP	Above XY	Above XY
Third	Below HP, behind VP	Below XY	Above XY
Fourth	Below HP, in front of VP	Below XY	Below XY

Projection of lines in first quadrant

Parallel to both HP and VP

Both views show true length and are parallel to XY.

Perpendicular to HP

Top view is a point; front view shows true length.

Perpendicular to VP

Front view is a point; top view shows true length.

Inclined to HP, parallel to VP

Front view shows true length and true inclination with HP.

Inclined to VP, parallel to HP

Top view shows true length and true inclination with VP.

Inclined to both planes

Neither view directly shows true length; use rotation or auxiliary method.

Solved example: Point A 30 mm above HP and 40 mm in front of VP

1. Draw XY reference line.
2. Since A is above HP, mark front view a′ 30 mm above XY.
3. Since A is in front of VP, mark top view a 40 mm below XY.
4. Draw a projector perpendicular to XY passing through a′ and a.
5. Label correctly.

Final result: a′ above XY and a below XY. Point is in first quadrant.

Solved example: Line AB parallel to both HP and VP

Given AB = 60 mm, 25 mm above HP, 35 mm in front of VP.

1. Draw XY line.
2. Draw front view a′b′ = 60 mm parallel to XY at 25 mm above XY.
3. From a′ and b′ draw projectors perpendicular to XY.
4. Draw top view ab = 60 mm parallel to XY at 35 mm below XY.
5. Ensure a is below a′ and b below b′.

Final result: both views show true length.

Common Mistake: Do not interchange front view and top view. In first quadrant, FV is above XY and TV is below XY.

MODULE 3 • CO3

Orthographic Projections and Sectional Views

Syllabus focus: orthographic, isometric and oblique concepts; first and third angle methods and symbols; conversion of pictorial views using first angle projection; full and half sections.

Module Outcomes

- M3.01 Summarize orthographic projections.
- M3.02 Develop sectional views.

Orthographic projection

Represents a 3D object using separate 2D views such as front view, top view and side view.

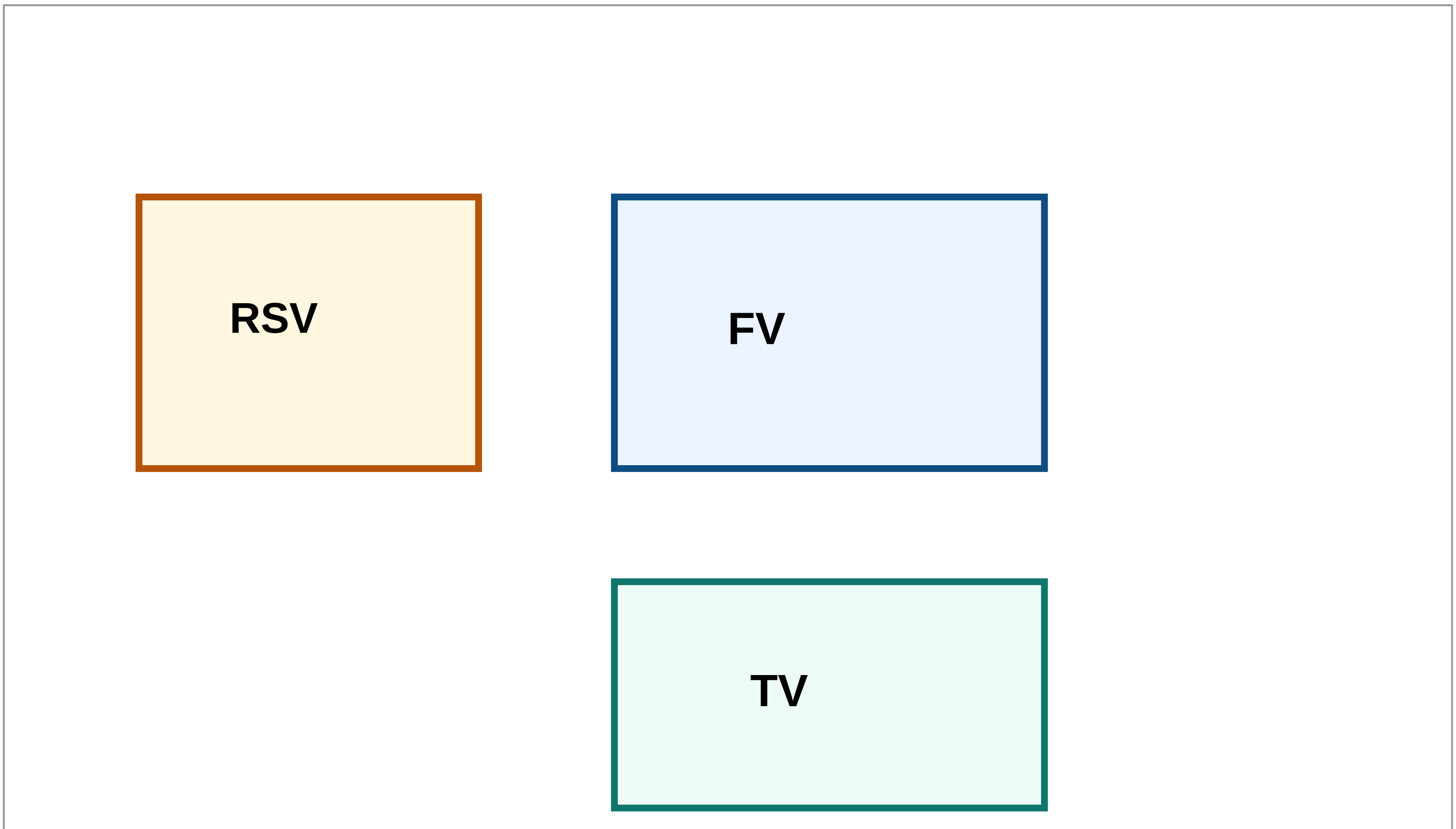
> Object-നെ front, top, side എന്നീ views ആയി കാണിക്കുന്നതാണ് orthographic projection.

First angle and third angle

Feature	First angle	Third angle
Top view	Below front view	Above front view
Right side view	Left of front view	Right of front view
Syllabus use	Use first angle only	Concept only

Drawing Rule: All drawings should be in first angle projection.

First angle layout



First angle: TV below FV, right side view on left

Orthographic procedure

1. Study pictorial view and identify height, width and depth.
2. Select front view showing maximum details.
3. Draw front view first.
4. Project down to draw top view.
5. Transfer depth to side view using mitre line if required.
6. Use thick, dashed and centre lines correctly.

Sectional views

Sectional drawings show internal details by imagining the object cut by a section plane. They reduce hidden lines and reveal holes, slots and internal cavities.

Object ഉള്ളിലെ details കാണിക്കാൻ മുറിച്ചതായി കരുതി വരയ്ക്കുന്ന view ആണ് section view.

Full section

Cutting plane passes completely through the object. Cut material is hatched.

Half section

Used for symmetrical objects: one half outside view, other half sectional interior.

Hatching rules

- Use thin equally spaced 45° lines.
- Hatch only cut material.
- Do not hatch holes or empty spaces.
- Adjacent parts use different spacing/direction.

Solved sectional view procedure

1. Study object and select section plane that reveals maximum internal details.
2. Mark cutting plane with arrows and label A-A.
3. Imagine front portion removed.
4. Draw visible edges of remaining object.
5. Hatch only cut solid areas using thin 45° lines.
6. Leave holes and empty spaces unhatched.
7. Add centre lines, dimensions and labels.

Common mistakes: missing cutting plane label, hatching holes, and unnecessary hidden lines.

MODULE 4 • CO4

Isometric Projections and CADD Introduction

Syllabus focus: isometric scale, isometric view/projection, conversion between orthographic and isometric, freehand sketching and CADD commands.

Module Outcomes

- M4.01 Develop isometric views of simple objects.
- M4.02 Convert orthographic views into isometric view/projection.
- M4.03 Prepare freehand sketches of real objects.
- M4.04 Identify CADD commands for 2D figures.

Isometric drawing

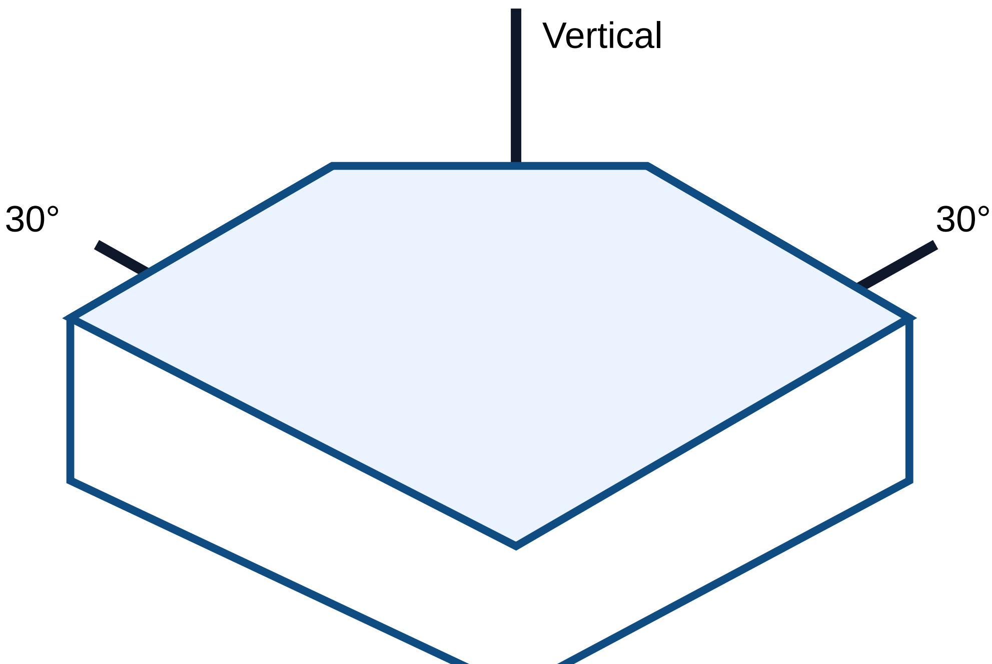
Represents a 3D object on a 2D sheet using three axes equally inclined. Vertical remains vertical; other axes are usually 30° to horizontal.

> Isometric drawing-ൽ object 3D പോലെ കാണിക്കും. vertical axis, 30° left, 30° right axes ഉപയോഗിക്കുന്നു.

Isometric view vs projection

Feature	Isometric View	Isometric Projection
Scale	True dimensions	Isometric scale
Size	Slightly larger	About 0.816 of true length
Use	When view is asked	When projection is asked

Isometric axes



Isometric axes: 120° between axes

Isometric procedure

1. Read orthographic views to get height, width and depth.
2. Draw isometric axes: vertical, 30° left and 30° right.

3. Construct an isometric box.
4. Mark steps, slots, holes or slant faces on correct planes.
5. Darken visible edges and remove construction lines.

CADD command bank

Command	Purpose
LINE, CIRCLE, ARC, RECTANGLE, POLYGON	Basic drawing commands
OFFSET, TRIM, EXTEND	Parallel copy, remove, lengthen entities
MOVE, COPY, ROTATE, MIRROR	Modify position and orientation
FILLET, CHAMFER	Round or bevel corners
DIMENSION, LAYER	Annotation and organisation
ZOOM, PAN, SAVE, PRINT/PLOT	View and output commands

CADD workflow: rectangle with circle

1. Start new drawing and set units to millimetres.
2. Use RECTANGLE command and specify corners.
3. Use CIRCLE command and locate centre point.
4. Use OFFSET if inner boundaries are needed.
5. Use TRIM to remove unwanted parts.
6. Use DIMENSION to add values.
7. Use LAYER for object, centre and dimension lines if needed.
8. SAVE and PRINT/PLOT.

DRAWING BANKS

Standards, construction, projection and command banks

Drawing Standards

Instruments, line types, lettering, BIS dimensioning, arrowheads, extension lines, first angle rules, hatching and isometric axes.

Construction Procedures

Regular polygons, ellipse, parabola, helix, involute, projections, orthographic views, sections, isometric and freehand sketching.

Projection Rules

HP, VP, XY, quadrants, FV/TV placement, first/third angle, true length, apparent length, HP/VP inclinations.

Dimension & Scale

Dimension line, extension line, arrowhead, leader line, chain, parallel, coordinate, plain, diagonal and vernier scales.

CADD

Purpose, benefits, draw commands, modify commands, view commands, dimension commands, layers, saving and plotting.

Exam Procedure

Given data, assumptions, construction steps, final drawing description, labels, projection method and mistakes to avoid.

PRACTICE DRAWING SECTION

Draw these problems on sheets

Module 1

1. Construct a hexagon side 35 mm.
2. Construct ellipse 100 × 60 by concentric circle method.
3. Construct parabola by tangent method.
4. Construct involute of circle diameter 50 mm.

Module 2

1. Point 35 mm above HP and 25 mm in front of VP.
2. Point 20 mm below HP and 30 mm behind VP.
3. Line 70 mm parallel to both planes.
4. Line inclined to HP and parallel to VP.

Module 3

1. Orthographic views of rectangular block with slot.
2. Views of object with cylindrical surface.
3. Full sectional view of a simple bush.
4. Half section of a symmetrical object.

Module 4

1. Isometric view of 60 × 40 × 30 block.
2. Isometric view of L-block.
3. Stepped block from orthographic views.
4. CADD steps for rectangle with two holes.

Check your drawing

- ✓ Correct measurements.
- ✓ Correct projection method.
- ✓ Different line types.
- ✓ Labels and dimensions readable.
- ✓ Final curve/view darkened.

Self-assessment

- ✓ I can place views around XY.
- ✓ I can construct conics and involute.
- ✓ I can draw first angle orthographic views.
- ✓ I can hatch sections correctly.
- ✓ I can draw isometric views.

✓ I can identify CADD commands.

QUICK REVISION

Last-day drawing revision

Module 1

- Line types must differ.
- Dimensions outside object.
- Curves smooth through points.

Module 2

- HP gives TV.
- VP gives FV.
- XY is reference line.
- Parallel plane shows true length.

Module 3

- Use first angle only.
- TV below FV.
- Hatch cut material only.

Module 4

- Axes: vertical and two 30°.
- View uses true length.
- Projection uses isometric scale.
- Know CADD commands.

Last Day Tip: Draw one curve, one point projection, one line projection, one orthographic view, one sectional view and one isometric view.

MODEL QUESTION PAPER WITH FULL ANSWERS

ENGINEERING GRAPHICS • Maximum Marks: 75 • Time: 3 Hours

Missing data, if any, may be suitably assumed. Sketches are to be accompanied wherever required. All drawings should be in first angle projection. This is a fresh model paper based on syllabus and official question paper format.

PART A - Answer all questions. Each carries 1 mark. [5 × 1 = 5]

1. What is the purpose of a centre line?
2. Name the plane on which top view is obtained.
3. State the position of top view in first angle projection.
4. Name any two CADD modify commands.
5. Sketch description: draw the three isometric axes.

PART B - Answer any five. Each carries 8 marks. [5 × 8 = 40]

1. Construct a regular hexagon of side 40 mm and write procedure.
2. Construct an ellipse 100 × 60 by concentric circle method.
3. Draw projections of point A, 30 mm above HP and 40 mm in front of VP.
4. Line AB is 60 mm long, parallel to both HP and VP, 25 mm above HP and 35 mm in front of VP. Draw projections.
5. Explain first angle projection and draw layout of FV, TV and right side view.
6. Write procedure for full sectional view of simple object with central hole.
7. Write CADD command sequence for rectangle 80 × 50 mm with centre circle radius 15 mm.

PART C - Answer any two. Each carries 15 marks. [2 × 15 = 30]

1. Construct involute of a circle diameter 50 mm with procedure.
2. Draw orthographic views of a simple stepped block in first angle projection. Assume dimensions.
3. Convert assumed orthographic data into isometric view: block 70 × 45 × 35 mm with 25 mm wide top step of height 15 mm.

ANSWER KEY / COMPLETE SOLUTION PROCEDURES

Full answers for the model paper

Part A answers

No.	Answer
1	A centre line indicates centre, axis of symmetry or path of rotation. It is a chain thin line.
2	Top view is obtained on HP, the Horizontal Plane.
3	In first angle projection, top view is placed below the front view.
4	TRIM, EXTEND, MOVE, COPY, ROTATE, MIRROR, OFFSET, FILLET, CHAMFER.
5	Draw one vertical axis and two axes at 30° to horizontal; axes are 120° apart.

Part B full solution procedures

B1. Regular hexagon side 40 mm

1. Draw AB = 40 mm.
2. With A and B as centres and radius 40 mm, draw arcs meeting at O.
3. Draw a circle with O as centre and radius OA.
4. Step off the same radius around circle to locate A, B, C, D, E, F.
5. Join adjacent points with continuous thick lines.
6. Label vertices and keep construction arcs light.

Expected final view: regular hexagon with all sides 40 mm.

B2. Ellipse by concentric circle method

Given major axis = 100 mm, minor axis = 60 mm.

1. Draw AB = 100 mm.
2. Bisect at O and draw CD = 60 mm perpendicular to AB.
3. Draw outer circle radius 50 mm and inner circle radius 30 mm.
4. Divide both circles into 12 equal parts.
5. From outer circle divisions draw vertical projectors.
6. From corresponding inner circle divisions draw horizontal projectors.
7. Intersections give ellipse points.
8. Join points smoothly and darken final ellipse.

Common mistake: do not join by straight segments.

B3. Projection of point A

1. Draw XY reference line.
2. Mark a′ 30 mm above XY because point is above HP.
3. Mark a 40 mm below XY because point is in front of VP.
4. Draw common projector perpendicular to XY.
5. Label views. Final placement: a′ above XY and a below XY.

B4. Projection of line parallel to both planes

Given AB = 60 mm, 25 mm above HP, 35 mm in front of VP.

1. Draw XY.
2. Draw a′b′ = 60 mm parallel to XY at 25 mm above XY.
3. Draw projectors from a′ and b′.
4. Draw ab = 60 mm parallel to XY at 35 mm below XY.
5. Ensure a below a′ and b below b′.
6. Darken and label. Both views show true length.

B5. First angle projection

1. In first angle projection, object is between observer and plane.
2. Draw front view at centre.
3. Place top view below front view.
4. Place right side view to the left of front view.
5. Use thin projectors for alignment.
6. Mention that this paper requires first angle projection.

Common mistake: placing right side view on the right side.

B6. Full sectional view of object with central hole

1. Select cutting plane through centre of the hole.
2. Mark cutting plane line with arrows and label A-A.
3. Imagine front half removed.
4. Draw remaining visible outline.
5. Draw hole as empty space.
6. Hatch only cut solid material with thin 45° lines.
7. Add centre lines, dimensions and labels.

Expected result: internal hole visible; cut material hatched.

B7. CADD command sequence

1. Set units to millimetres.
2. RECTANGLE: specify first corner 0,0 and opposite corner 80,50.
3. Centre of rectangle is 40,25.
4. CIRCLE: centre 40,25 and radius 15.
5. DIMENSION: add length 80, height 50 and radius 15.
6. Use LAYER if needed.
7. SAVE and PRINT/PLOT after checking line weights.

Part C full solution procedures

C1. Involute of circle diameter 50 mm

1. Draw circle diameter 50 mm, centre O.
2. Divide circle into 12 equal parts.
3. Draw tangents at each division point.
4. On tangent at point 1 set off one division length; at point 2 set off two division lengths; continue similarly.
5. Mark all obtained points.
6. Join points smoothly to form involute.
7. Darken final curve and keep tangents light.
8. Label base circle, division points and involute.

Common mistakes: wrong tangent direction, unequal divisions and sharp curve joints.

C2. Orthographic views of stepped block

Assume base block length 80 mm, depth 50 mm, total height 50 mm. Step length 35 mm and step height 25 mm on top left.

1. Draw front view first because step profile is clear.
2. Draw outer rectangle 80 × 50.
3. Mark step: 35 mm from left and 25 mm height difference.
4. Use thick visible lines.
5. Project downward to draw top view below FV using first angle projection.
6. Set depth 50 mm in top view.
7. Draw right side view on left side of FV.
8. Show height and depth in side view.
9. Add dimensions and labels.

Expected layout: top view below front view and right side view on left.

C3. Isometric view of stepped block

Given block 70 × 45 × 35 mm with 25 mm wide top step of height 15 mm.
1. Draw isometric axes: vertical and two 30° axes.
2. Construct enclosing isometric box 70 length, 45 depth, 35 height using true dimensions.
3. Mark the top step on top face with width 25 mm.
4. Mark step height 15 mm downward from top surface.
5. Draw visible vertical and inclined step edges.
6. Remove unnecessary construction lines.
7. Darken visible final edges.
8. Do not show hidden lines unless asked.
9. Check all receding lines are parallel to 30° axes.

Expected final drawing: neat isometric stepped block with correct proportions.

MCQ Answer Key

1. B - HP. 2. B - Below front view. 3. C - Chain thin. 4. B - TRIM. 5. C - 120°.

REFERENCES AND ONLINE RESOURCES

Official learning resources

Code	Book Title / Author
T1	P. I. Varghese, Engineering Graphics, VIP Publishers
T2	K. C. John, Engineering Graphics, PHI Learning Private Limited
T3	K. N. Anilkumar, Engineering Graphics unique methods, easy solutions, Adhyuth Narayan Publishers, Kottayam
R1	N. D. Bhatt, Engineering Drawing, Charotar Publishing House, Anand, Gujarat, 2010; ISBN: 978-93-80358-17-8
R2	D. M. Kulkarni, A. P. Rastogi, A. K. Sarkar, Engineering Graphics with AutoCAD, PHI Learning Private Limited, New Delhi, 2010; ISBN: 978-8120337831
R3	M. B. Shah and B. C. Rana, Engineering Drawing, Pearson Publications
R4	T. Jayapoovan, Engineering Drawing & Graphics using AutoCAD, Vikas Publications

Online Resources: NPTEL Engineering Drawing, Engineering Graphics video playlists and CADD/isometric drawing resources listed in the official syllabus.

FINAL DRAWING EXAM CHECKLIST

Check before submitting your sheet

Drawing quality

- ✓ Sheet is clean.
- ✓ Construction lines are light.
- ✓ Final lines are dark and clear.
- ✓ Labels and dimensions are readable.

Technical correctness

- ✓ First angle projection is followed.
- ✓ XY and projectors are correct.
- ✓ Top/side view placement is correct.
- ✓ Curves are smooth.
- ✓ Hatching is only on cut material.
- ✓ Isometric axes are correct.

Final Reminder: Engineering Graphics is scored through visible procedure. Keep construction logical, neat, labelled and readable.